

Ahmed Ikram | Software Engineering

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PROFILE

3rd-year BSc Computer Science (Data Science & AI) student at the University of Dundee, on track for a First-Class degree. Builds production-ready systems end-to-end - from React and React Native frontends through to secure Flask/FastAPI REST APIs, real-time Socket.IO layers, and cloud infrastructure on both AWS and Azure, with 1,452 automated tests enforcing 85% line coverage across every PR and a Terraform-provisioned Azure estate of 30+ resources deployed via OIDC-federated CI/CD with zero static credentials. Experienced leading a 7-person Agile team, independently architecting a 10-service Dockerised platform, and diagnosing distributed system bottlenecks - including an 8-9x JDBC performance fix on a live Spark pipeline. Seeking a Software Engineering or Full-Stack post-graduation role to contribute across the entire stack in a team that ships production software.

EDUCATION

High School of Dundee

Aug 2017 - June 2023

- National 5 Computer Science, Maths, English, Economics, Physics, Chemistry, Biology (A)
- Higher Computer Science, Maths and English (A), Higher Chemistry (B) and Higher Physics (C)
- Advanced Higher Computer Science (C)

University of Dundee - BSc (Hons) Computer Science (Data Science and AI)

Sep 2023 - Present

- Expected Graduation: June 2027

PROJECTS

DevSync Project Management Platform with GitHub Integration | React, Tailwind, Python, Flask, PostgreSQL, Socket.io, GitHub OAuth, AWS, Docker, GitHub Actions, CI/CD | Jan 2025 - Apr 2026

- React, Tailwind frontend, Flask & Gunicorn REST API over a 12-table indexed PostgreSQL schema, and a JWT-authenticated Socket.IO layer with project-scoped rooms - live task updates, notifications, and dashboard refreshes broadcast only to members viewing the affected project, with no page refresh required; GitHub activity fetched via GraphQL API to reduce latency when creating charts and reports
- Implemented three-tier RBAC (Developer / Team Leader / Admin) via role decorators on every route; JWT in HTTP-only cookies; GitHub OAuth 2.0 handled server-side with tokens never sent to the browser; bidirectional task ↔ issues/PR sync means closing a GitHub issue automatically updates the linked task in DevSync
- Production-like AWS infrastructure: ECS Fargate behind an ALB, RDS in a private subnet, and S3, CloudFront & ACM for the React SPA - frontend deployment blocked until ECS health checks pass to prevent API/UI version mismatch; 1,452 automated tests at 85%/85% coverage hard-gate every PR before any AWS step runs, OIDC federation for CI/CD with no static credentials, secrets injected at runtime via AWS Secrets Manager and per-task IAM roles scoped to least-privilege

ATM Log Aggregation, Analysis & Diagnostics Platform | Python, FastAPI, React, PostgreSQL, Kafka, Redis, nginx, Docker, AWS, MLflow | Mar 2026 - June 2026

- Led backend and data engineering on a 7-person Agile team delivering a diagnostics platform for NCR Atleos; independently architected an Apache Kafka event-driven pipeline, 3-layer ML detection engine, Agentic RAG assistant, and full 10-service Docker infrastructure, significantly extending the team-delivered project
- Architected a 10-service Dockerised system with health-check cascading startup (PostgreSQL → Kafka → services → frontend): FastAPI REST API (30 endpoints, 6 routers, JWT + Redis blacklist auth), 8 Redis patterns (rate limiting, distributed lock, Pub/Sub, DLQ, HyperLogLog, caching, dedup, JWT blacklist), and a React 19 frontend via multi-stage Docker build (~25MB nginx-alpine final image) with environment-parity proxy rewrites replacing the Vite dev server's '/api/*' routing in production nginx
- Validated the full system via 670 tests across 10 tiers (unit, integration, ML, RAG, Kafka, stress, security) with an isolated test PostgreSQL DB; all anomaly detection runs versioned in MLflow v3 on AWS (RDS + S3)

StockLens FinTech Spending & Investment App | React Native Expo, TypeScript, Node.js, PostgreSQL |

Oct 2025 - Dec 2025

- Built a full-stack mobile app in React Native/TypeScript with secure biometric authentication, real-time Firebase synchronisation, and custom AES encryption for sensitive data at rest
- Integrated a Node.js OCR pipeline parsing physical receipts into structured JSON records, querying the Alpha Vantage API to map spending to stock tickers and compute historical CAGR - covered by 78 Jest tests across OCR fallbacks, API caching, cryptography roundtrips, auth flows and more
- Engineered an ML layer combining ARIMA-based time-series forecasting for per-category spending trends and Linear Regression for forward-looking portfolio projections, fed by historical CAGR computed from Alpha Vantage data - surfaced directly in the app UI

PROJECTS

W3C Web Logs ETL Pipeline | *Apache Airflow, Python, Databricks, dbt, Apache Spark, PySpark, Azure SQL, Terraform, Docker, GitHub Actions, Power BI, Grafana, Prometheus* | Feb 2026 - June 2026

- Built a serverless Databricks medallion pipeline (Bronze → Silver → Azure SQL → dbt → Power BI) ingesting 93 IIS log files (155,380 rows, 0 dropped) across 7 @dlt.expect_or_drop quality gates; a custom UDF handles 14- vs 18-field format detection dynamically; 4 Airflow DAGs orchestrate the full pipeline with zero data-processing logic in DAG files - layers decoupled via Airflow Datasets, no ExternalTaskSensor
- Diagnosed and resolved an 8-9× JDBC export bottleneck (413s → 45s) by isolating three root causes: Spark-side pre-filtering before collect(), tuple(row) over asDict() eliminating 4.7M dict allocations, and removing a redundant .count() scan
- Provisioned 30+ Azure resources across 2 Terraform configurations including the full OIDC auth chain (Azure AD app, service principal, federated credential, role assignment) via a single terraform apply; 7-workflow CI/CD pipeline with 4 parallel jobs and a rollback workflow that re-applies Terraform from the previous commit on demand — zero static credentials anywhere
- Delivered a 7-page Power BI dashboard (156K requests, 88 countries, P95 1,149ms, 38% bot traffic, 9.7% 404 rate) from 18 dbt-generated CSV exports via a semantic contract; observability: custom Python freshness probe → 3 Grafana dashboards (23 panels) → 10 alert rules across 3 severity-tiered action groups; 613 tests across 5 tiers including dual-dialect dbt CI against side-by-side PostgreSQL and SQL Server containers

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, SQL, Bash
Frontend	React, React Native, Tailwind, Vite, Chart.js
Backend	FastAPI, Flask, Node.js, Apache Airflow, dbt, GraphQL API, Socket.IO, Kafka, Gunicorn
Security	JWT, OAuth 2.0, RBAC, HTTP-only Cookies, OIDC Federation, AWS Secrets Manager
Databases	PostgreSQL, SQLite, MySQL, MongoDB, Redis
Cloud & DevOps	AWS (ECS Fargate, EC2, S3, RDS, IAM, CloudFront, Sagemaker, Secrets Manager), Azure (Databricks, Azure SQL, ADLS), Terraform, Docker, nginx, GitHub Actions, Linux
Observability	Grafana, Prometheus, StatsD
Testing	Pytest, Jest, React Testing Library, Cypress, dbt Tests

CERTIFICATES

AWS Academy Graduate - Machine Learning Foundations

Feb 2025

AWS Academy Graduate - Cloud Foundations

Mar 2024

Foundations of Azure AI: Concepts, Capabilities, and Implementation

Oct 2024

EXPERIENCE

Beaulys Mini Market, Dundee, Scotland

Retail Assistant

September 2023 - Present

- Troubleshoot and resolved Point-of-Sale and stock system errors quickly to minimise customer wait time

Zulus Peri Peri Grill, Dundee, Scotland

Front of House Worker

June 2021 - September 2023

- Trained 5 new staff members, helping them get up to speed quickly during busy service periods

TRANSFERRABLE SKILLS

Agile & Iterative Delivery

Comfortable owning work end-to-end within sprint cycles - scoping, building, testing, and shipping incrementally while adapting to changing requirements

System Design & Scalability

Approach problems by reasoning about scalability, security, and maintainability from the start - not retrofitted after the fact

End-to-End Ownership

Take full responsibility for delivery From requirements and user stories through schema design, API documentation, and architecture decisions through to cloud deployment and post-release monitoring against formal acceptance criteria

Test-Driven Development & Code Quality

Enforce quality gates via automated test suites (Pytest, Jest, Cypress) at 80–90% line coverage; treat test coverage as a non-negotiable, not an afterthought

Technical Communication

Explain API behaviour, architecture trade-offs, and system design decisions clearly to both technical peers and non-technical stakeholders

Handling Ambiguity

Adept at breaking loosely defined problems into concrete technical tasks with clear acceptance criteria and measurable outcomes

